AI and Coding Workshop

Instructor: Karin Huangsuwan
3000747 Healthcare Innovation and Entrepreneurship

AI is Getting Better Every Day

1997

Optimization

Focused on solving closed-system puzzles. AI could calculate millions of permutations to find the "best" move in games like Chess.

2012

Identifying Images

Neural networks allowed AI to "see." Machines began to categorize the world, identifying objects and faces with superhuman accuracy.

2017

Understanding Language

AI moved beyond keywords to grasp semantics. It began to understand the nuances of human intent, sentiment, and complex instructions.

2022

Writing and Generating

Generative models turned data into expression. AI became capable of drafting professional copy and rendering photorealistic imagery from scratch.

2023

Reasoning

Models began to "think" through problems. This involved chain-of-thought processing to solve math, debug code, and justify its own decisions.

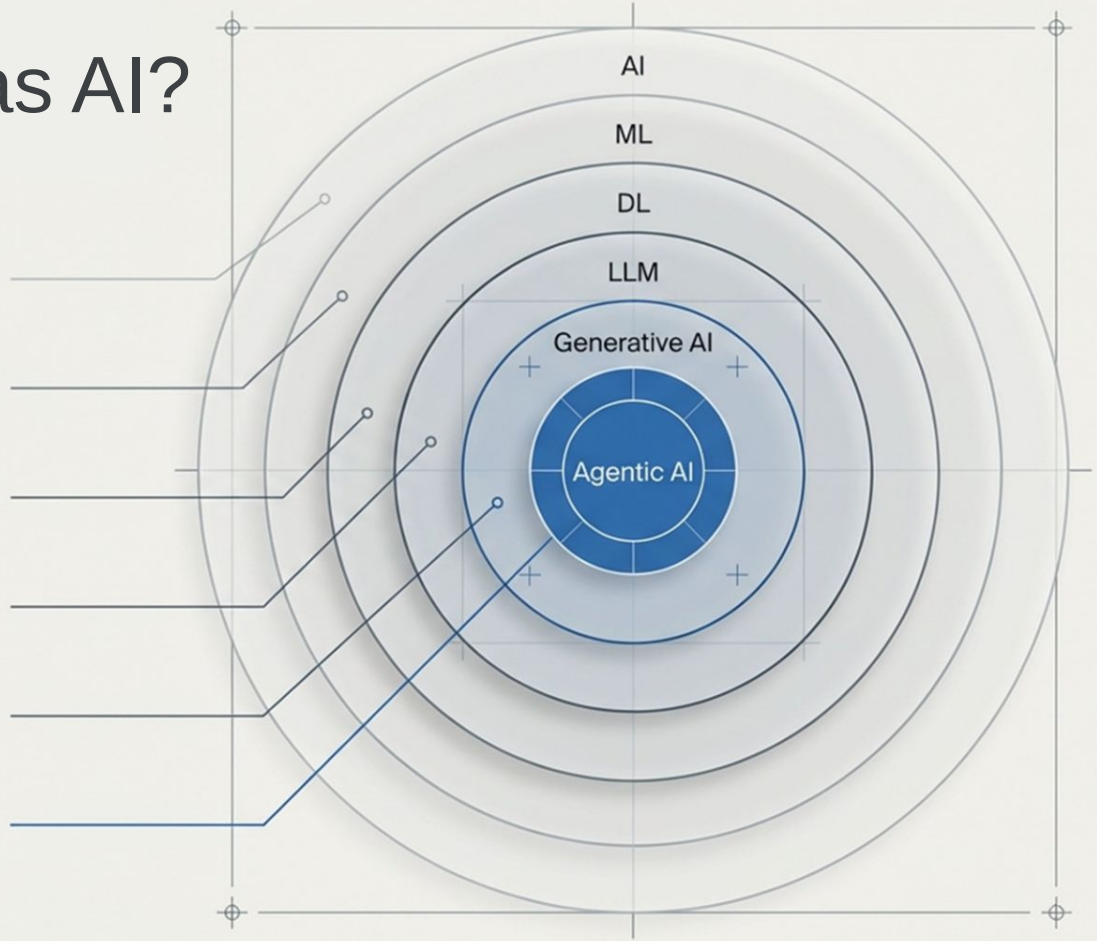
Today

Autonomous Action

AI now executes multi-step workflows. It doesn't just suggest a plan; it uses tools, browses the web, and completes the task independently.

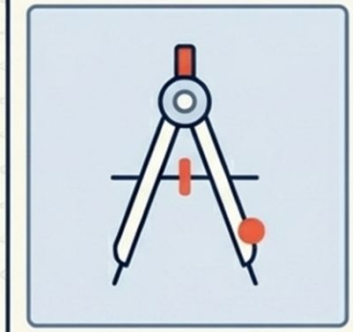
What is classified as AI?

Artificial Intelligence: The broad field of making machines smart (e.g., a chess engine).
Machine Learning: Learning patterns from data instead of explicit programming.
Deep Learning: Neural networks powering modern image recognition and translation.
Large Language Model: Massive text training to predict the next word in a sequence.
Generative AI: Creating new content rather than classifying existing data.
Agentic AI: Using tools and making decisions to achieve a goal. Acting, not just talking.



NotebookLM

Before Vibe Coding: Who Designs an AI App?



Software Architect

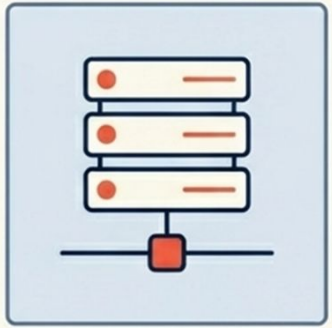
The Urban Planner

Domain: Tech stack, databases, and cross-service roadmaps.

Software Engineer

The Builder

Domain: Algorithms, data structures, and feature implementation.



DevOps / SRE

The Grid Operator

Domain: Server infrastructure, uptime, and traffic management.

Product Manager

The Visionary

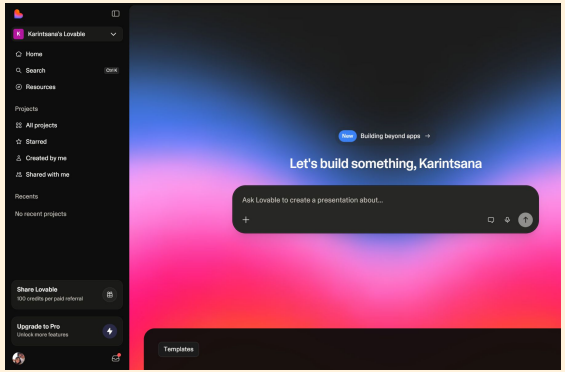
Domain: User needs and ensuring the architecture serves a human purpose.

NotebookLM

After Vibe Coding: Code in English (or Any Language)

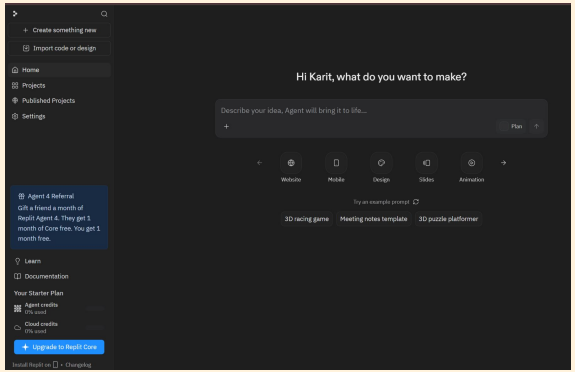
Vibe coding is a software development approach where

- you build applications primarily through natural language conversation with an AI
- rather than writing code line-by-line yourself.



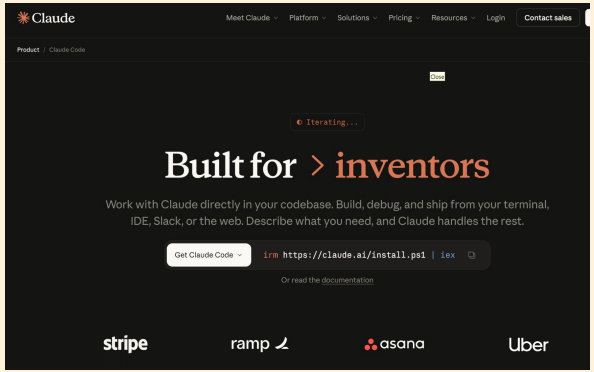
Lovable: The Visual Prototyper

Best for non-coders and rapid MVPs. It "vibes" high-fidelity UI and auto-provisions **Supabase** backends from simple text or screenshots.



Replit: The All-in-One Studio

The cloud-native powerhouse for **full-stack** apps. Its agent handles the code, database, and hosting in a single browser tab.



Claude Code: The Terminal Agent

The developer's expert for existing projects. It operates directly on your **local files** to refactor, debug, and run tests via command line.

How to Vibe Code?

1. **Prompting:** You describe what you want in plain English (e.g., *"Make a dashboard that tracks my stock portfolio and turns the background red if I'm losing money"*).
2. **Generation:** An AI tool (like **Cursor**, **Replit Agent**, or **Claude**) writes the entire block of code for you.
3. **Iterating:** You run the app, see what’s wrong, and "talk" the AI through the fix (e.g., *"The red is too bright, make it a soft pastel and add a 'Refresh' button"*).
4. **Testing:** Automated bug-finding and validation to ensure the app actually works.
5. **Deployment:** deployment to a live environment, moving from "thought" to "product."

Current Advanced AI Tools

1. Large Language Models (LLM)
2. Computer Vision Models
3. Machine Learning Models

Large Language Model: A system that has "read" almost everything on the internet.

ChatGPT

Gemini

Claude

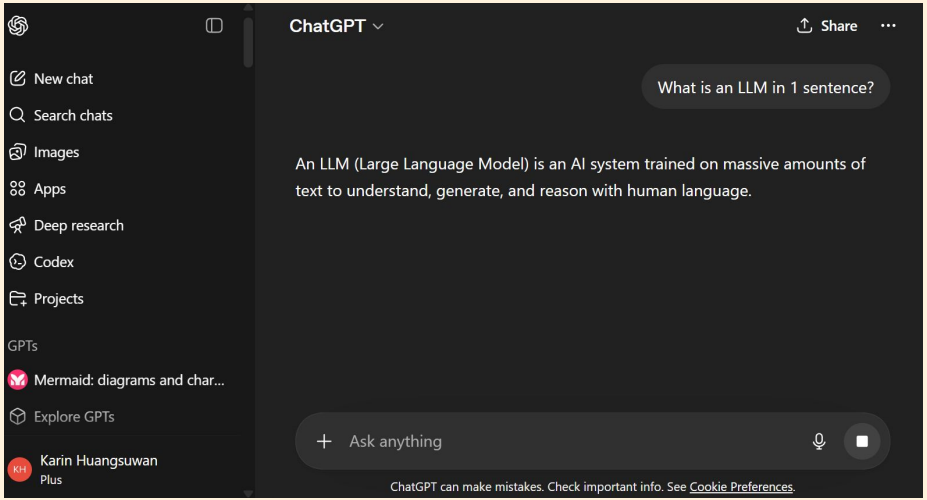
Accessing Large Language Models

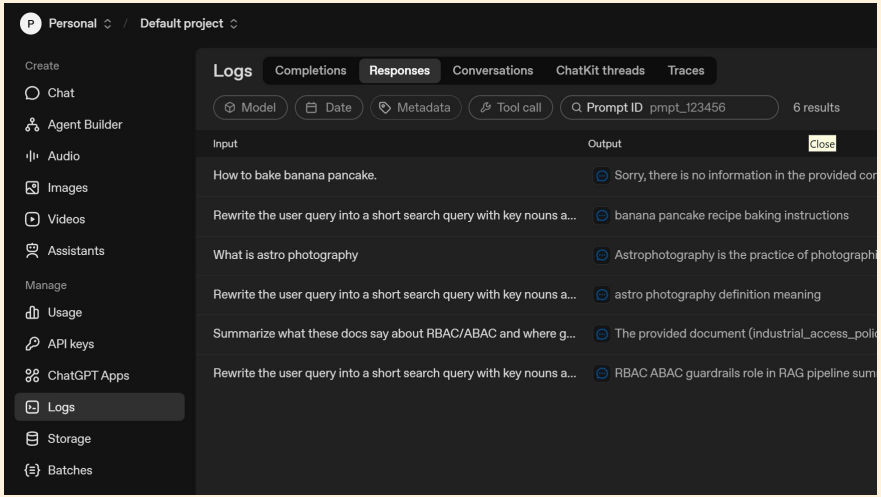
1. Web UI, good for: everyday use

https://chatgpt.com/
https://gemini.google.com/app
https://claude.ai

2. Developer API, good for: app development that involves LLM calling

http://platform.openai.com/
https://aistudio.google.com/
https://platform.claude.com/





RAG System: Add Knowledge to LLM

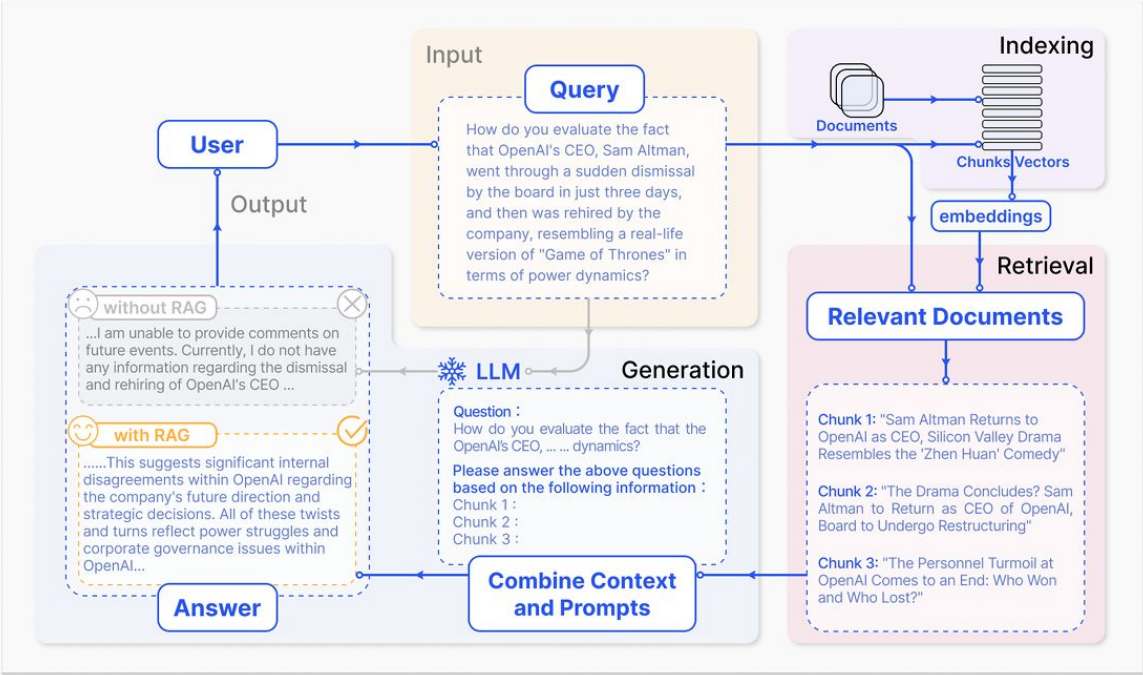
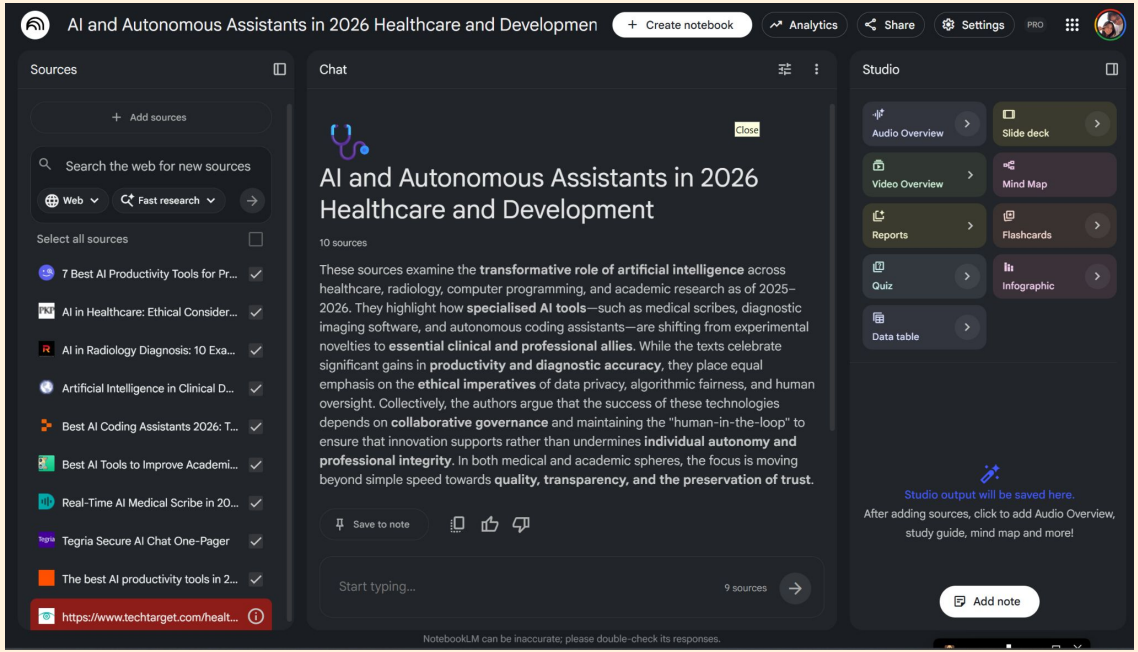


Fig. 2. A representative instance of the RAG process applied to question answering. It mainly consists of 3 steps. 1) Indexing. Documents are split into chunks, encoded into vectors, and stored in a vector database. 2) Retrieval. Retrieve the Top k chunks most relevant to the question based on semantic similarity. 3) Generation. Input the original question and the retrieved chunks together into LLM to generate the final answer.

NotebookLM: Research Helper

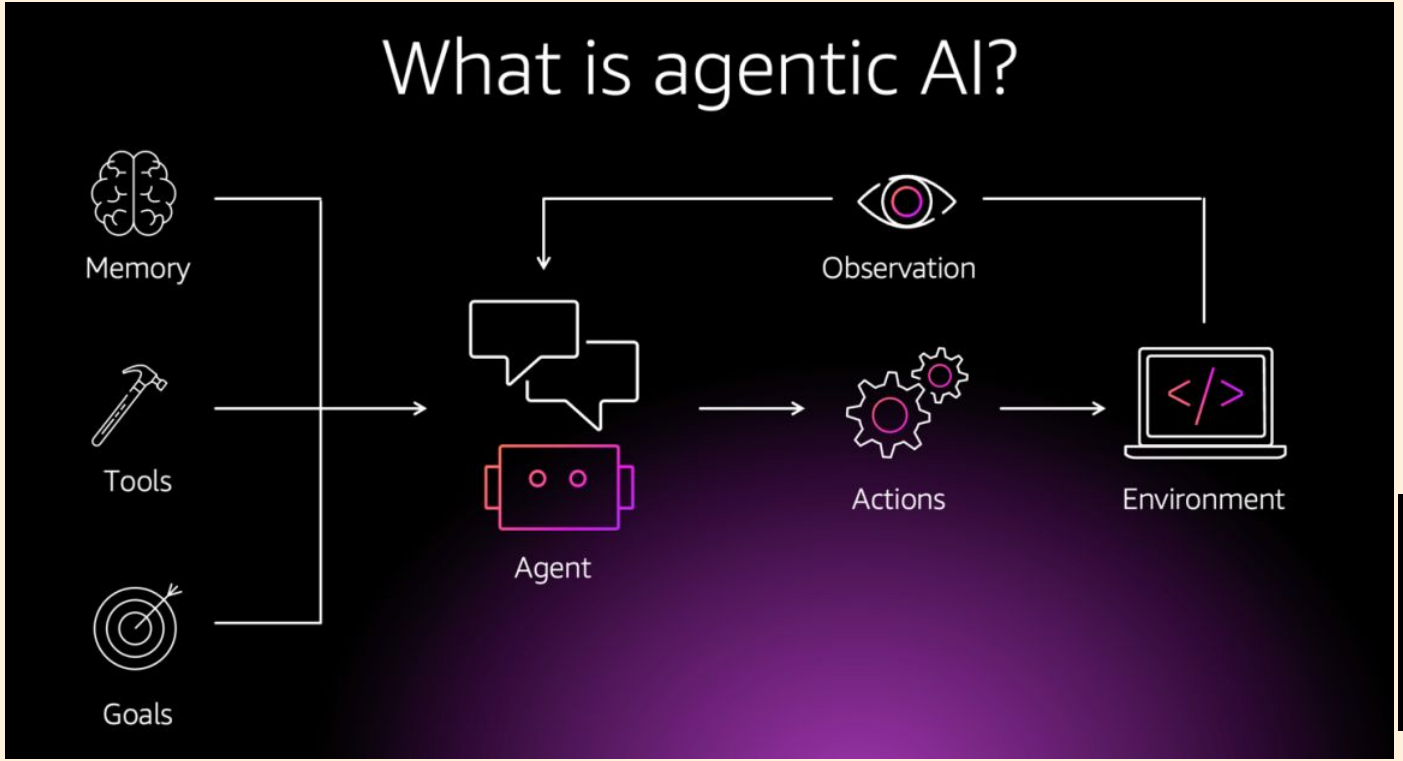


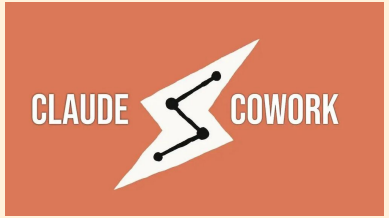
Core Capabilities:

- **Sources Panel:** Upload and manage your data. It supports a massive **1 million token context window** per notebook, allowing you to analyze dozens of documents, entire books, or hours of video simultaneously.
- **Chat Panel:** Ask questions, summarize complex topics, or brainstorm ideas specifically based on your sources. It provides inline citations so you can verify exactly where the information came from.
- **Studio Panel:** Instantly transform your research into "artifacts" like reports, slide decks, and data tables.

https://notebooklm.google.com/

Agentic AI: The Future of Workflow Automation

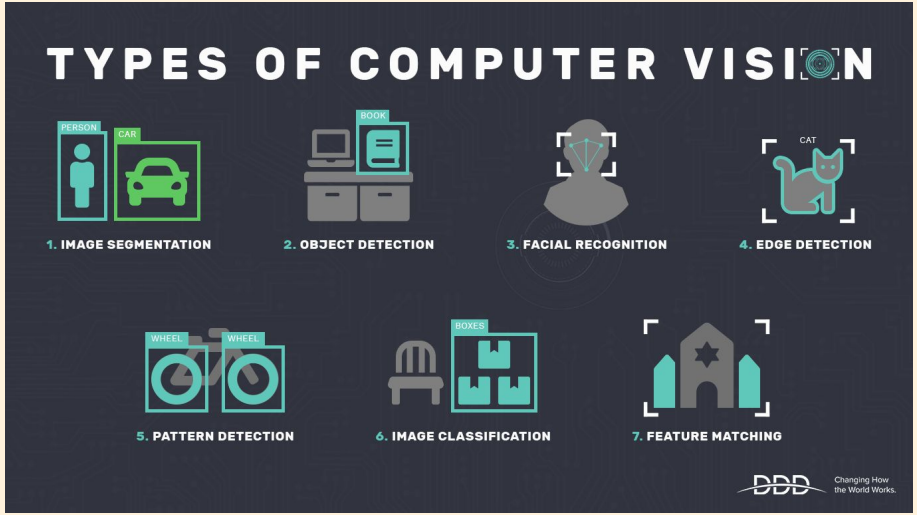




https://aws.amazon.com/blogs/machine-learning/build-agentic-systems-with-crewai-and-amazon-bedrock/

Deep Learning Models

Computer Vision Models: AI systems designed to interpret and understand visual data from the world, such as images and videos, by identifying patterns and features.



https://www.digitaldividedata.com/blog/everything-about-computer-vision

Open-source hubs for computer vision models:

- **Hugging Face (Computer Vision Hub):** huggingface.co/models?pipeline_tag=image-classification
  - *Best for:* The widest variety of modern models (Transformers, Diffusion, etc.) and easy integration with Python.
- **PyTorch Hub:** pytorch.org/hub
  - *Best for:* Research-ready models that can be loaded with a single line of code (`torch.hub.load`).
- **TensorFlow Hub (now Kaggle Models):** kaggle.com/models/google/resnet-v2
  - *Best for:* Models optimized for production and mobile deployment (TensorFlow Lite).
- **Papers with Code:** paperswithcode.com/area/computer-vision
  - *Best for:* Finding the "State of the Art" (SOTA) for specific tasks like object detection or segmentation, linked directly to their GitHub code.
- **Model Zoo (OpenCV):** github.com/opencv/opencv/wiki/Model-Zoo
  - *Best for:* Finding lightweight models specifically optimized to run fast on CPUs or edge devices.

Integrating Open-Source CV Models into Your App

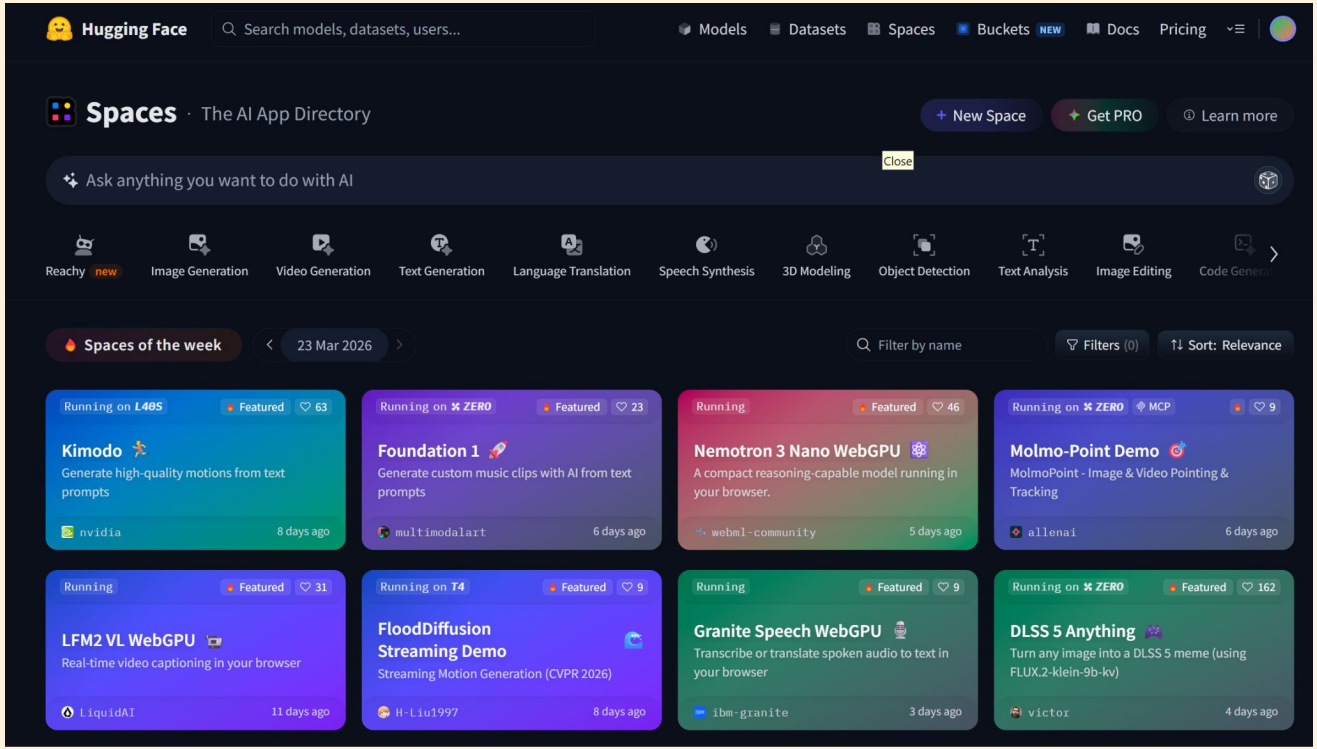
1. Choose your hosting environment (where the model will run on)
   a. Cloud Hosting (Server-Side): The model runs on a powerful remote server (GPU-enabled). Your app sends an image via an API and gets a result back.
   b. Edge/On-Device Hosting: The model is converted and compressed to run directly on the user’s phone or computer.
2. Prepare the Model for Hosting

- Pickling/Saving: Saving the weights (e.g., .pth for PyTorch, .h5 for TensorFlow).

3. Setting up the API

- Containerization: Use Docker to package the model, its libraries (OpenCV, PyTorch), and the code into one portable "container."
- Web Frameworks: Wrap the model in a FastAPI or Flask script. This creates a URL (e.g., https://api.myapp.com/predict) that your app can "POST" images to.

Free Hosting Environment: Huggingface Spaces



A **Hugging Face Space** is a free, hosted environment that allows you to turn your AI model into a live, interactive web application or API in minutes.

https://huggingface.co/spaces